

Section 2: Hands-on UI-Based Questions

1. S3 Bucket Configuration

איך להפעיל גירסאות (enable versioning):
כאשר יוצרים S3, יש לעבור ללשונית Properties.
יש לגלול מטה ל-Bucket Versioning ← ללחוץ על Edit.
לבחור Enable ולאחר מכן ללחוץ על שמור שינויים.

The screenshot shows the Amazon S3 console interface. The left sidebar contains navigation links for 'Amazon S3', 'General purpose buckets', 'Directory buckets', 'Table buckets', 'Access Grants', 'Access Points', 'Object Lambda Access Points', 'Multi-Region Access Points', 'Batch Operations', 'IAM Access Analyzer for S3', 'Block Public Access settings for this account', 'Storage Lens', 'Dashboards', 'Storage Lens groups', and 'AWS Organizations settings'. The main content area is titled 'omerasus-bucket' and has tabs for 'Objects', 'Metadata', 'Properties' (selected), 'Permissions', 'Metrics', 'Management', and 'Access Points'. Under the 'Properties' tab, there is a 'Bucket overview' section with fields for 'AWS Region' (US East (N. Virginia) us-east-1), 'Amazon Resource Name (ARN)' (arn:aws:s3:::omerasus-bucket), and 'Creation date' (February 7, 2025, 10:41:44 (UTC+02:00)). Below this is the 'Bucket Versioning' section, which shows 'Bucket Versioning' is 'Enabled' and includes an 'Edit' button. A description explains that versioning keeps multiple variants of an object. The 'Multi-factor authentication (MFA) delete' section shows it is 'Disabled' and includes a 'Learn more' link.

הפוליסה של הדלי שיצרתי, מצרף בנוסף את הקוד במצב עריכה כי לא הצלחתי להכניס את כל הקוד לתמונה.

The screenshot shows the Amazon S3 console interface, specifically the 'Bucket policy' tab for the 'omerasus-bucket'. The left sidebar is the same as in the previous screenshot. The main content area shows 'Block all public access' is 'On'. Below this, there is a 'Bucket policy' section with 'Edit' and 'Delete' buttons. A message states: 'Public access is blocked because Block Public Access settings are turned on for this bucket'. The JSON policy is displayed in a text area with a 'Copy' button. The policy allows the user 'OmerAsus' to perform 's3:ListBucket' and 's3:GetObject' actions on the bucket 'omerasus-bucket'.

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "AWS": "arn:aws:iam::504949722475:user/OmerAsus"
8       },
9       "Action": "s3:ListBucket",
10      "Resource": "arn:aws:s3:::omerasus-bucket"
11    },
12    {
13      "Effect": "Allow",
14      "Principal": {
15        "AWS": "arn:aws:iam::504949722475:user/OmerAsus"
16      },
17      "Action": [
18        "s3:GetObject",
19        "s3:PutObject",
20        "s3:DeleteObject"
21      ],
22      "Resource": "arn:aws:s3:::omerasus-bucket/*"
23    }
24  ]
25 }

```

2. Launch an EC2 Instance

מצורף המכונה שיצרתי והפוליסה שמראה שיש את ה2 הפורטים הנדרשים.

Instance summary for i-0dc8509e07cd8ff48 (OmerAsus-ec2) [info](#)

Updated less than a minute ago

[Connect](#)
[Instance state](#)
[Actions](#)

Instance ID

i-0dc8509e07cd8ff48

Public IPv4 address

98.81.221.67 [open address](#)

Private IPv4 addresses

172.31.80.19

IPv6 address

-

Instance state

Running

Public IPv4 DNS

ec2-98-81-221-67.compute-1.amazonaws.com [open address](#)

Hostname type

IP name: ip-172-31-80-19.ec2.internal

Private IP DNS name (IPv4 only)

ip-172-31-80-19.ec2.internal

Elastic IP addresses

-

Answer private resource DNS name

IPv4 (A)

Instance type

t2.micro

AWS Compute Optimizer finding

[Opt-in to AWS Compute Optimizer for recommendations.](#) [Learn more](#)

Auto-assigned IP address

98.81.221.67 [Public IP]

VPC ID

vpc-0c3424b7237e51c20 (jessica-vpc) [open](#)

Auto Scaling Group name

-

IAM Role

-

Subnet ID

subnet-0040f9cf696fbdc73 [open](#)

Managed

false

IMDSv2

Required

Instance ARN

arn:aws:ec2:us-east-1:504949722475:instance/i-0dc8509e07cd8ff48

Operator

-

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM Role

-

Owner ID

504949722475

Launch time

Fri Feb 07 2025 12:11:21 GMT+0200 (Israel Standard Time)

Security groups

sg-0daac789233921122 (OmerAsus-sg) [open](#)

sg-0daac789233921122 - OmerAsus-sg [Actions](#)

Details

Security group name

OmerAsus-sg

Security group ID

sg-0daac789233921122

Description

omer asus security group

VPC ID

vpc-0c3424b7237e51c20 [open](#)

Owner

504949722475

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (2) [Manage tags](#) [Edit inbound rules](#)

Search

☐

Name

▼

Security group rule ID

▼

IP version

▼

Type

▼

Protocol

▼

Port range

▼

Source

▼

Description

▼

☐

-

sg-r-07f756793ca184a80

IPv4

SSH

TCP

22

0.0.0.0/0

-

☐

-

sg-r-0caedc49c6349db18

IPv4

HTTP

TCP

80

0.0.0.0/0

-

3. Configure an IAM User with S3 Access

The screenshot shows the AWS IAM console for the user **OmerAsus**. The **Summary** tab is active, displaying the user's ARN (`arn:aws:iam::504949722475:user/OmerAsus`), console access status (Disabled), and two access keys. The **Permissions** tab is also visible, showing a single policy named **OmerAsus-S3_policy** attached directly to the user. The policy details are shown in a code editor, defining permissions for the `arn:aws:s3::omerasus-bucket`.

```
1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Sid": "Statement1",
6-       "Effect": "Allow",
7-       "Action": [
8-         "s3:*"
9-       ],
10-      "Resource": [
11-        "arn:aws:s3::omerasus-bucket"
12-      ]
13-    }
14-  ]
15- }
```

בשביל לוודא שיש למשתמש שלי את ההרשאות הנכונות אני אעלה ממנו קובץ לדלי שיצרתי ויש לו הרשאות אליו -

תמונה מהטרמינל של ההעלאת הקובץ דרך היוזר שלי-

```
asus@h-MacBook-Pro-s1-wmr aws % aws s3 cp test-file.txt s3://omerasus-bucket/ --profile OmerAsus
upload: ./test-file.txt to s3://omerasus-bucket/test-file.txt
asus@h-MacBook-Pro-s1-wmr aws % aws s3 ls s3://omerasus-bucket/ --profile OmerAsus

2025-02-07 12:34:52      40 test-file.txt
asus@h-MacBook-Pro-s1-wmr aws %
```

תמונה מהדלי עצמו-

The screenshot shows the AWS S3 console for the bucket **omerasus-bucket**. The **Objects** tab is active, displaying a list of objects. One object, **test-file.txt**, is listed with a size of 40.0 B and a storage class of Standard. The console also shows options for managing the bucket, such as creating a folder or uploading new objects.

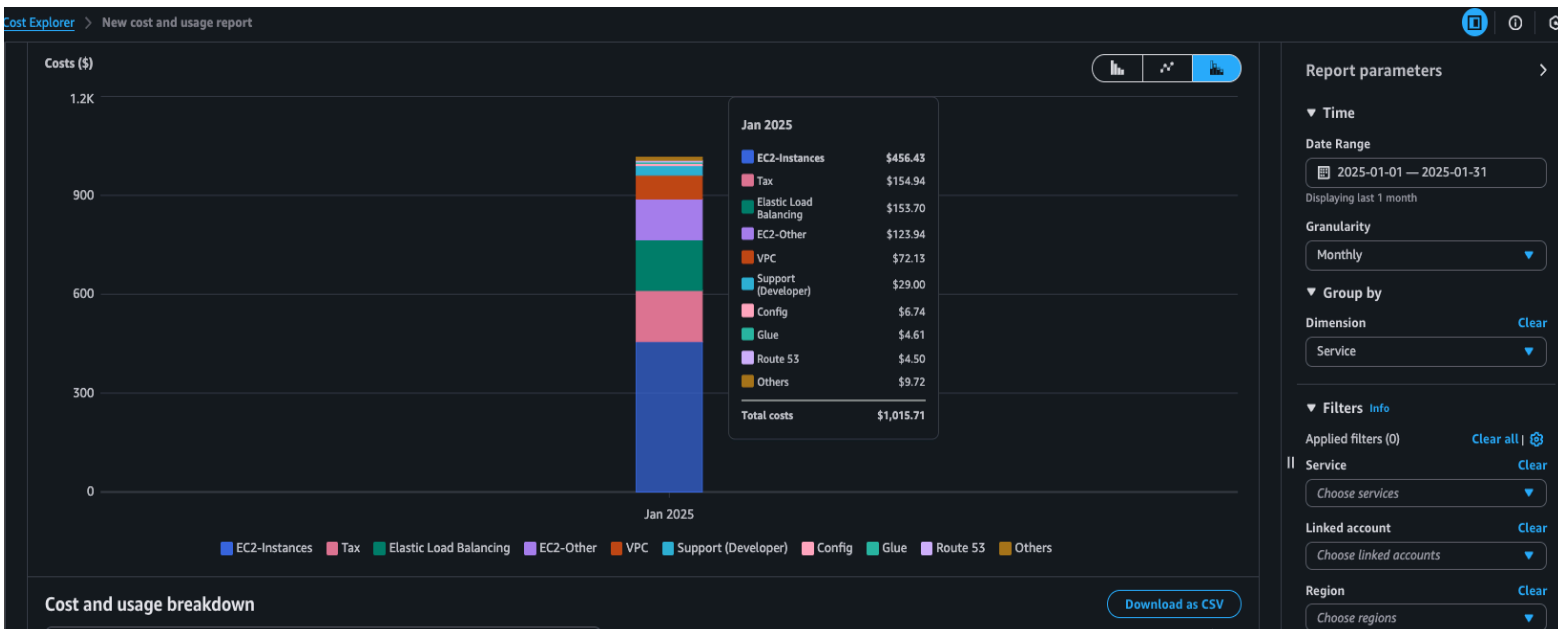
4. Set Up a CloudWatch Alarm

כאשר יגיע לקצה ויצטרך לשלוח התראה, יצרתי sns חדש שנקרא omer-noti-bymail אשר שולח למייל שלי בעת הצורך התראה.



5. Identify AWS Billing Costs

איך לנתח עלויות ב-aws:
ראשית עבור אל AWS Cost Explorer
תלחץ על Billing and Pricing Management ב-AWS Console ופתח את Cost Explorer.
בחר את תקופת החיוב מצד ימין, במקרה שלנו - חודש אחרון.
לאחר מכן תוכל לראות במספר אופציות את הכלים בהם השתמשת וכמה הוצאת על כל אחד (למשל על ec2 הוציאו בחודש האחרון - \$456)



Section 3: Hands-on advanced

1. Deploy an Auto Scaling Group with a Single EC2 Instance

omerasus-autoscale

omerasus-autoscale Capacity overview

arn:aws:autoscaling:us-east-1:504949722475:autoScalingGroup:c22da6a4-af06-428b-a7b9-9bd336c10f41:autoScalingGroupName/omerasus-autoscale

Desired capacity	Scaling limits (Min - Max)	Desired capacity type	Status
1	1 - 1	Units (number of instances)	-

Date created
Fri Feb 07 2025 13:49:06 GMT+0200 (Israel Standard Time)

Details

Integrations - new

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Launch template

Launch template

lt-0815229673403498e

OmerAsus--autoscale

Version

Default

Description

-

View details in the launch template console

AMI ID

ami-04681163a08179f28

Security groups

-

Storage (volumes)

-

Instance type

t2.micro

Security group IDs

sg-0daac789233921122

Key pair name

omerasus-key

Owner

am:aws:iam::504949722475:user/jb-user

Create time

Fri Feb 07 2025 13:38:49 GMT+0200 (Israel Standard Time)

Request Spot Instances

No

omerasus-autoscale

omerasus-autoscale Capacity overview

arn:aws:autoscaling:us-east-1:504949722475:autoScalingGroup:c22da6a4-af06-428b-a7b9-9bd336c10f41:autoScalingGroupName/omerasus-autoscale

Desired capacity	Scaling limits (Min - Max)	Desired capacity type	Status
1	1 - 1	Units (number of instances)	-

Date created
Fri Feb 07 2025 13:49:06 GMT+0200 (Israel Standard Time)

Details

Integrations - new

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Load balancing

Load balancer target groups

OmerAsus-TG

Classic Load Balancers

-

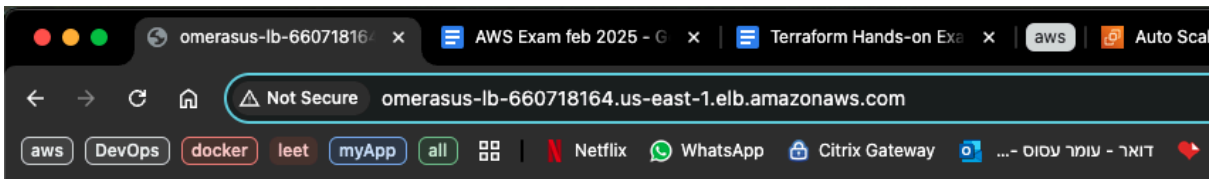
2. Connect to the EC2 Instance and Install Nginx

בתמונה המצורפת ניתן לראות את הסטטוס של nginx (ע"י הפקודה `systemctl status nginx`) ומתחת את ההדפסה (ע"י הפקודה `curl http://localhost:80`)

```
[ec2-user@ip-172-31-92-28 ~]$ systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; vendor preset: disabled)
   Active: active (running) since Fri 2025-02-07 12:00:20 UTC; 2min 23s ago
     Main PID: 3481 (nginx)
    CGroup: /system.slice/nginx.service
            └─3481 nginx: master process /usr/sbin/nginx
               └─3482 nginx: worker process

Feb 07 12:00:20 ip-172-31-92-28.ec2.internal systemd[1]: Starting The nginx HTTP and reverse pr.....
Feb 07 12:00:20 ip-172-31-92-28.ec2.internal nginx[3475]: nginx: the configuration file /etc/ng...ok
Feb 07 12:00:20 ip-172-31-92-28.ec2.internal nginx[3475]: nginx: configuration file /etc/nginx/...ul
Feb 07 12:00:20 ip-172-31-92-28.ec2.internal systemd[1]: Started The nginx HTTP and reverse pro...r.
Hint: Some lines were ellipsized, use -l to show in full.
[ec2-user@ip-172-31-92-28 ~]$ curl http://localhost:80
<h1>Welcome to AWS Auto Scaling</h1>
[ec2-user@ip-172-31-92-28 ~]$
```

3. Access the Web Page via the Load Balancer



Welcome to AWS Auto Scaling

4. IAM User Setup for S3 Access

כאן ניתן לראות את היוזר שיוצר - OmerAsus

OmerAsus

Summary

ARN
arn:aws:iam::504949722475:user/OmerAsus

Created
February 07, 2025, 10:53 (UTC+02:00)

Console access
Disabled

Last console sign-in
-

Access key 1
AKIA... - Active
Never used. Created today.

Access key 2
Create access key

Permissions

Groups

Tags

Security credentials

Last Accessed

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type
All types

Search

Policy name

Type

Attached via

OmerAsus-S3_policy

Customer managed

Directly

OmerAsus-S3_policy

Copy JSON

Edit

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Statement1",
      "Effect": "Allow",
      "Action": [
        "s3:*"
      ],
      "Resource": [
        "arn:aws:s3:::omerasus-bucket"
      ]
    }
  ]
}
```

כאן ניתן לראות כיצד אני מעלה קובץ לדלי שמשוייך אליו ורואה לאחר מכן את הקובץ שהעליתי בנוסף לקובץ אחר שהעליתי מקודם -

```
aws — ec2-user@ip-172-31-92-28:~ — zsh — 100x26
[asus@h-MacBook-Pro-s1-wmr aws % aws s3 cp another-test.txt s3://omerasus-bucket/ --profile OmerAsus ]
upload: ./another-test.txt to s3://omerasus-bucket/another-test.txt
[asus@h-MacBook-Pro-s1-wmr aws % aws s3 ls s3://omerasus-bucket/ --profile OmerAsus
2025-02-07 14:24:08      40 another-test.txt
2025-02-07 12:34:52      40 test-file.txt
[asus@h-MacBook-Pro-s1-wmr aws % ]
```

omerasus-bucket

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

Show versions

1

Name	Type	Last modified	Size	Storage class
another-test.txt	txt	February 7, 2025, 14:24:08 (UTC+02:00)	40.0 B	Standard
test-file.txt	txt	February 7, 2025, 12:34:52 (UTC+02:00)	40.0 B	Standard

5. Create a CloudWatch Alarm for CPU Usage

